

International Journal of Theoretical and Applied Finance
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INSTRUCTIONS FOR TYPESETTING MANUSCRIPTS USING L^AT_EX*

FIRST AUTHOR[†]

*University Department, University Name, Address
City, State ZIP/Zone, Country[‡]
author_id@domain_name[§]*

SECOND AUTHOR

*Group, Laboratory, Address
City, State ZIP/Zone, Country
author_id@domain_name*

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Keywords: Keyword1; keyword2; keyword3.

1. General Appearance

Contributions to the *International Journal of Theoretical and Applied Finance* will mostly be processed by using the authors' source files. These should be submitted with the manuscripts, and resubmitted in the final form if a paper requires revision before being accepted for publication.

2. The Main Text

Authors are encouraged to have their contribution checked for grammar. American spelling should be used. Abbreviations are allowed but should be spelled out in

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full when first used. Integers ten and below are to be spelled out. Italicize foreign language phrases (e.g. Latin, French).

The text should be in 10 pt roman, single spaced with baselineskip of 13 pt. Text area (excluding copyright block and folio) is 6.9 inches high and 5 inches wide for the first page. Text area (excluding running title) is 7.7 inches high and 5 inches wide for subsequent pages. Final pagination and insertion of running titles will be done by the publisher.

3. Major Headings

Major headings should be typeset in boldface with the first letter of important words capitalized.

3.1. *Sub-headings*

Sub-headings should be typeset in boldface italic and capitalize the first letter of the first word only. Section number to be in boldface roman.

3.1.1. *Sub-subheadings*

Typeset sub-subheadings in medium face italic and capitalize the first letter of the first word only. Section numbers to be in roman.

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Sections, sub-sections and sub-subsections are numbered in Arabic. Use double spacing before all section headings, and single spacing after section headings. Flush left all paragraphs that follow after section headings.

3.3. *Lists of items*

List may be presented with each item marked by bullets and numbers.

Bulleted items

- item one,
- item two,
- item three.

Numbered items

1. item one,
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3. item three.

The order of subdivisions of items in bullet and numbered lists may be presented as follows:

Bulleted items

- First item in the first level
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 - * First item in the third level
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 - Third item in the second level
 - Fourth item in the second level
- Third item in the first level
- Fourth item in the first level

Numbered items

1. First item in the first level
2. Second item in the first level
 - (a) First item in the second level
 - (b) Second item in the second level
 - i. First item in the third level
 - ii. Second item in the third level
 - iii. Third item in the third level
 - (c) Third item in the second level
 - (d) Fourth item in the second level
3. Third item in the first level
4. Fourth item in the first level

4. Equations

Displayed equations should be numbered consecutively, with the number set flush right and enclosed in parentheses. The equation numbers should be consecutive within each section. Numbering by sections refers only to the large main sections, i.e. to those with Level 1 headings. Thus the equations in Sec. 4 would be labeled (4.1), (4.2), etc.

$$\mu(n, t) = \frac{\sum_{i=1}^{\infty} 1(d_i < t, N(d_i) = n)}{\int_{\sigma=0}^t 1(N(\sigma) = n) d\sigma}. \quad (4.1)$$

Equations should be referred to in abbreviated form, e.g. “Eq. (4.1)” or “(4.2)”. In multiple-line equations, the number should be given on the last line.

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For multiline equations, prefer using `align` or `align*` instead of `eqnarray`.

$$\begin{aligned}
 g(x) &= \sum_{k=1}^N y_k g(x_k) \\
 &= \sum_{k=1}^N y_k \sum_{l=1}^M b_{kl} w_l \\
 &= \sum_{k=1}^N \sum_{l=1}^M b_{kl} y_k w_l
 \end{aligned} \tag{4.2}$$

Displayed equations are to be centered on the page width. Standard English letters like x are to appear as x (italicized) in the text if they are used as mathematical symbols. Punctuation marks are used at the end of equations as if they appeared directly in the text.

5. Theorem Environments

Theorem 5.1. *Theorems, lemmas, definitions, etc. are set on a separate paragraph, with extra 1 line space above and below. They are to be numbered consecutively within each section.*

$$\mu(n, t) = \frac{\sum_{i=1}^{\infty} 1(d_i < t, N(d_i) = n)}{\int_{\sigma=0}^t 1(N(\sigma) = n) d\sigma}. \tag{5.1}$$

Numbering by sections refers only to the large main sections, i.e. to those with Level 1 headings. Thus the numbering style in Sec. 5 would be labeled (5.1), (5.2), etc.

The theorem environments text can be cross-linked to the body text, e.g., Theorem 5.1 and Lemma 5.1.

Lemma 5.1 (Sample). *Theorems, lemmas, definitions, etc. are set on a separate paragraph, with extra 1 line space above and below. They are to be numbered consecutively within each section.*

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Proof. Proofs should end with a box. □

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Figures are to be inserted in the text nearest their first reference. Figure placements can be either top or bottom. If the author requires the publisher to reduce the figures, ensure that the figures (including letterings and numbers) are large enough to be clearly seen after reduction. If photographs are to be used, only black and white ones are acceptable.

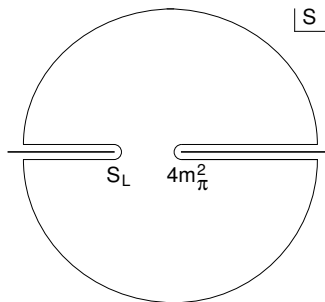


Fig. 1. A schematic illustration of dissociative recombination. The direct mechanism, $4m_\pi^2$ is initiated when the molecular ion S_L captures an electron with kinetic energy.

Figures are to be sequentially numbered in Arabic numerals. The caption must be placed below the figure. Typeset in 8 pt roman with baselineskip of 10 pt. Long captions are to be justified by the “page-width”. Use double spacing between a caption and the text that follows immediately.

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Tables should be inserted in the text as close to the point of reference as possible. Some space should be left above and below the table.

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If tables need to extend over to a second page, the continuation of the table should be preceded by a caption, e.g. “Table 2. (*Continued*)”. Notes to tables are placed below the final row of the table and should be flushleft. Footnotes in tables shouldbe indicated by superscript lowercase letters and placed beneath the table.

Table 1. Comparison of acoustic for frequencies for piston-cylinder problem.

Piston mass	Analytical frequency (Rad/s)	TRIA6- S_1 model (Rad/s)	% Error
1.0	281.0	280.81	0.07
0.1	876.0	875.74	0.03
0.01	2441.0	2441.0	0.0
0.001	4130.0	4129.3	0.16

Note: Table notes
^aTable footnote A
^bTable footnote B

8. Running Heads

Please provide a shortened runninghead (not more than eight words) for the title of your paper. This will appear on the top right-hand side of your paper.

9. Footnotes

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Appendix A. Appendices

Appendices should be used only when absolutely necessary. They should come before the Acknowledgments. If there is more than one appendix, number them alphabetically. Number displayed equations occurring in the Appendix in this way, e.g. (A.1), (A.2), etc.

$$\mu(n, t) = \frac{\sum_{i=1}^{\infty} 1(d_i < t, N(d_i) = n)}{\int_{\sigma=0}^t 1(N(\sigma) = n) d\sigma}. \quad (\text{A.1})$$

Acknowledgments

This section should come before the References. Funding information may also be included here.

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References are to be listed in alphabetical order using the last name of the first-named author. They are to be cited in the text in an author-date format (i.e. Black 1973). They can be typed after punctuation marks, e.g. "... in the statement (Duffie 2001) ..." or used directly, e.g. "see Duffie (2001) for examples." For multi-authored papers, please use the '&' symbol, i.e. Delbaen & Schachermayer (1994).

Please list the references using the style shown in the following examples:

^aFootnotes should be typeset in 8 pt roman at the bottom of the page.

Journal articles:

R. C. Merton (1974) On the pricing of corporate debt: the risk structure of interest rates, *Journal of Finance* **29** (2), 449–470, <https://doi.org/10.2307/2978814>.

F. Delbaen & W. Schachermayer (1994) A general version of the fundamental theorem of asset pricing, *Mathematische Annalen* **300** (1), 463–520, <https://doi.org/10.1007/BF01450498>.

D. C. Brody, L. P. Hughston & A. Macrina (2008a) Information-based asset pricing, *International Journal of Theoretical and Applied Finance* **11** (1), 107–142, <https://doi.org/10.1142/S0219024908004749>.

D. C. Brody, L. P. Hughston & A. Macrina (2008b) Dam rain and cumulative gain, *Proceedings of the Royal Society A* **464**, 1801–1822, <https://doi.org/10.1098/rspa.2007.0273>.

Notes:

1. List articles alphabetically by the surname of the first-named author.
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Books:

D. Duffie (2001) *Dynamic Asset Pricing Theory*, third edition. Princeton, New Jersey: Princeton University Press.

J. Gatheral (2006) *The Implied Volatility Surface: A Practitioner's Guide*. New York: Wiley.

R. A. Jarrow (2008) *Financial Derivatives Pricing: Selected Works of Robert Jarrow*. New Jersey: World Scientific Publishing Company, <https://doi.org/10.1142/6911>.

I. Karatzas & S. E. Shreve (1998) *Methods of Mathematical Finance*. New York: Springer-Verlag, <https://doi.org/10.1007/978-1-4939-6845-9>.

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2. Generally, avoid abbreviations, except in particular situations where the type-setting can be improved by use of an abbreviation (e.g., to avoid running over a line or a page).
3. Thus, write “third edition” rather than “3rd ed.”. However, “ed.” and “eds.” can be used for “editor” and “editors”.

Articles appearing in edited books:

J. Gatheral, A. Schied & A. Slynko (2011) Exponential resilience and decay of market impact. In: *Econophysics of Order-Driven Markets* (F. Abergel, B. K. Chakrabarti, A. Chakraborti & M. Mitra, eds.), 225–236. Milan: Springer, https://doi.org/10.1007/978-88-470-1766-5_15.

D. C. Brody, L. P. Hughston & A. Macrina (2007) Beyond hazard rates: a new framework for credit-risk modeling. In: *Advances in Mathematical Finance* (M. C. Fu, R. Jarrow, J. Y. Yen & R. Elliot, eds.), 231–257. Boston: Birkhauser, https://doi.org/10.1007/978-0-8176-4545-8_13.

PhD theses:

A. Macrina (2006) *An Information-Based Framework for Asset Pricing: X-Factor Theory and its Applications*. PhD Thesis, Department of Mathematics, King’s College London.

Articles appearing on ArXiv, but unpublished, or not yet accepted for publication:

R. A. Jarrow & M. Larsson (2014) Informational efficiency under short sale constraints. arXiv:1401.1851.

Example of reference list at end of article:

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- D. C. Brody, L. P. Hughston & A. Macrina (2007) Beyond hazard rates: a new framework for credit-risk modeling. In: *Advances in Mathematical Finance* (M. C. Fu, R. Jarrow, J. Y. Yen & R. Elliot, eds.), 231–257. Boston: Birkhauser, https://doi.org/10.1007/978-0-8176-4545-8_13.
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- D. C. Brody, L. P. Hughston & A. Macrina (2008b) Dam rain and cumulative gain, *Proceedings of the Royal Society A* **464**, 1801–1822, <https://doi.org/10.1098/rspa.2007.0273>.
- F. Delbaen & W. Schachermayer (1994) A general version of the fundamental theorem of asset pricing, *Mathematische Annalen* **300** (1), 463–520, <https://doi.org/10.1007/BF01450498>.

- D. Duffie (2001) *Dynamic Asset Pricing Theory*, third edition. Princeton, New Jersey: Princeton University Press.
- J. Gatheral (2006) *The Implied Volatility Surface: A Practitioner's Guide*. New York: Wiley.
- J. Gatheral, A. Schied & A. Slynko (2011) Exponential resilience and decay of market impact. In: *Econophysics of Order-Driven Markets* (F. Abergel, B. K Chakrabarti, A. Chakraborti & M. Mitra, eds.), 225–236. Milan: Springer, https://doi.org/10.1007/978-88-470-1766-5_15.
- R. A. Jarrow (2008) *Financial Derivatives Pricing: Selected Works of Robert Jarrow*. New Jersey: World Scientific Publishing Company, <https://doi.org/10.1142/6911>.
- R. A. Jarrow & M. Larsson (2014) Informational efficiency under short sale constraints. arXiv:1401.1851.
- I. Karatzas & S. E. Shreve (1998) *Methods of Mathematical Finance*. New York: Springer-Verlag, <https://doi.org/10.1007/978-1-4939-6845-9>.
- A. Macrina (2006) *An Information-Based Framework for Asset Pricing: X-Factor Theory and its Applications*. PhD Thesis, Department of Mathematics, King's College London.
- D. Madan & W. Schoutens (2011) Conic finance and the corporate balance sheet, *International Journal of Theoretical and Applied Finance* **14** (5) 587–610. Reprinted (2013) in: *Finance at Fields* (M. R. Grasselli & L. P. Hughston, eds.), 451–474, https://doi.org/10.1142/9789814407892_0020.
- R. C. Merton (1974) On the pricing of corporate debt: the risk structure of interest rates, *Journal of Finance* **29** (2), 449–470, <https://doi.org/10.2307/2978814>.

Note:

References are to be listed in alphabetical order using the last name of the first-named author.

Examples of citations:

- (a) This result was first established by Merton (1974).
- (b) This result is discussed in Merton (1974).
- (c) The arbitrage-free approach to interest rate theory has a long history (Merton 1974).
- (d) The theory of arbitrage free pricing has been greatly advanced in recent years (Delbaen & Schachermayer 1994, Karatzas & Shreve 1998, Duffie 2001).
- (e) See Delbaen & Schachermayer (1994), Karatzas & Shreve (1998), Duffie (2001), for further information.
- (f) The role of information in finance is generally appreciated (Brody *et al.* 2007, 2008a,b).
- (g) For more details about the role of information, see Brody *et al.* (2007, 2008a,b).